

AI and AI-Designed Interventions for Promoting Cooperation in Human Groups

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Summary

Many social problems, from climate change to resource allocation, require individuals to cooperate despite short-term incentives to defect. Recent work suggests that artificial intelligence can help groups overcome such collective action problems, either by designing social mechanisms that incentivise cooperation or by mediating disagreement among people with diverse preferences. However, it remains unclear when AI-designed interventions are effective, when humans accept them as legitimate, and whether their success depends on transparency, fairness, or perceived human agency.

This proposal asks whether AI-designed mechanisms can promote sustainable cooperation in repeated social dilemmas, and what psychological factors determine whether humans follow and endorse these mechanisms. I propose three linked studies: first, simulations to discover candidate AI-designed mechanisms; second, online behavioural experiments testing these mechanisms with human participants; and third, experiments manipulating transparency and democratic input to explain when AI interventions are trusted and accepted.

Background and Motivation

Human cooperation is central to many collective action problems. In repeated social dilemmas, individuals may benefit from free-riding in the short term, but groups achieve better outcomes when members contribute to shared resources or coordinate on mutually beneficial actions. Traditional approaches to promoting cooperation include punishment, reward, redistribution, reputation, communication, and institutional design. Yet designing mechanisms that are both effective and perceived as fair remains challenging, especially when individuals differ in their preferences, beliefs, and willingness to contribute.

Recent work in human-centred AI provides a promising new approach. Rather than using AI only to predict human behaviour, AI systems can be trained to design interventions that shape group outcomes. For example, “Democratic AI” uses reinforcement learning to discover redistribution mechanisms that humans prefer in incentivised economic games. Related work on common-pool resource problems shows that deep reinforcement learning can discover allocation policies that increase sustainable human cooperation. In parallel, work on AI-mediated deliberation suggests that language models can help people with diverse opinions identify common ground.

Despite these advances, an important gap remains. Existing work shows that AI can sometimes improve collective outcomes, but we still know relatively little about the psychological conditions under which humans accept AI-designed mechanisms. A mechanism that maximises welfare may fail if it is perceived as unfair, opaque, coercive, or misaligned with group norms. Conversely, a simpler or more transparent mechanism may achieve similar behavioural benefits while being more acceptable to human participants. Understanding this trade-off is crucial if AI is to be used to promote cooperation fairly and beneficially.

Research Question and Hypotheses

The central research question is:

Under what conditions can AI-designed mechanisms promote sustainable cooperation in human groups, and how do transparency, fairness, and perceived agency shape whether humans accept

and follow these mechanisms?

I will test three hypotheses.

H1: AI-designed mechanisms can improve cooperation. Mechanisms discovered through reinforcement learning will increase group-level cooperation and sustainability relative to baseline mechanisms such as equal allocation, proportional allocation, or no intervention.

H2: Effectiveness and acceptability can diverge. The mechanism that maximises aggregate welfare will not necessarily be the mechanism most preferred by humans. Participants' acceptance will depend on perceived fairness, transparency, and whether the mechanism is seen as respecting individual agency.

H3: Transparent or democratically constrained AI mechanisms will generalise better. AI-designed mechanisms that are interpretable, explainable, or subject to human choice will be more robust across groups and more likely to be followed over repeated interactions.

Methodological Overview

The project will combine computational modelling, reinforcement learning, and online behavioural experiments. Participants will interact in repeated social dilemmas, such as the prisoner's dilemma games, donation games, or common-pool resource games. These paradigms are well suited for studying cooperation because they create a tension between individual incentives and collective welfare.

The AI system will be treated as a social planner or mediator. Depending on the experiment, it may allocate resources, recommend contributions, impose rewards or sanctions, or generate explanations of group-level policies. Behavioural outcomes will include contribution rates, resource sustainability, inequality, group welfare, and defection rates. Psychological outcomes will include perceived fairness, trust in the AI, perceived autonomy, willingness to use the mechanism again, and explicit voting preferences.

Experiment 1: LLM Advice and Human Cooperation in Networked Social Dilemmas

Aim

Experiment 1 will test whether advice from large language models promotes or inhibits human cooperation in repeated social dilemmas. In everyday life, people increasingly use AI systems not as autonomous decision-makers, but as conversational advisors. This raises an important question: when individuals face a conflict between self-interest and collective welfare, does consulting an LLM make them more cooperative, or does it encourage strategic self-protection?

Design

The experiment will have two parts. In Experiment 1a, I will first characterise the behaviour of LLMs in repeated social dilemmas. LLM agents will play repeated donation games (or Prisoner's Dilemma) on a network. For each decision state, including the player's previous action, neighbours' previous actions, and payoff history, I will query the LLM offline repeatedly to estimate a stochastic advice strategy over cooperation and defection. This will allow me to measure whether the LLM's recommendations are unconditionally cooperative, conditionally cooperative, retaliatory, or self-interested. Furthermore, as a LLM baseline, I will use these strategies to conduct multi-round simulations to assess the final level of cooperation, etc.

In Experiment 1b, human participants will play a networked repeated social dilemma. Participants will independently make decision to cooperate or defect in each round as a No-advice baseline.

In Experiment 1c, human participants will play a networked repeated social dilemma. In each round, participants will first make an initial decision to cooperate or defect. They will then have the opportunity to consult

an LLM, which will recommend either cooperation or defection and provide a brief explanation. Participants will then make their final decision. This design allows me to compare initial and final decisions within the same round and to measure whether LLM advice changes human behaviour.

The main behavioural outcomes will be initial cooperation rate, final cooperation rate, advice adoption rate, switching from defection to cooperation, switching from cooperation to defection, and cumulative payoff. Psychological outcomes will include satisfaction with the advice, perceived fairness, trust in the AI, perceived autonomy, and willingness to consult the AI again.

Predictions

I predict that LLM advice will increase cooperation on average, especially in early rounds and in neighbourhoods where previous cooperation has been relatively high. However, I do not expect the LLM to recommend cooperation unconditionally. Instead, its advice may depend on social history: it may encourage cooperation when reciprocity is possible, but recommend defection or self-protection when a participant has repeatedly been exploited by neighbours.

I also predict that participants will selectively adopt LLM advice. They may be more likely to follow the LLM when its recommendation is consistent with their initial intention, when the explanation appeals to fairness or mutual benefit, or when previous AI advice has led to better outcomes. Conversely, participants may reject advice that appears too self-sacrificial, too selfish, or insufficiently sensitive to the network context. Some studies suggest that, given neutral prompts, LLM strategies may be closer to GRIM (cooperation in the first round; once a player defects, they will always defect thereafter). Humans may be less willing than LLMs to adopt unforgiving strategies, especially when such advice conflicts with social norms of forgiveness, leniency, or giving others a second chance.

This experiment will show whether AI can promote cooperation not by directly controlling incentives, but by shaping how individuals reason about cooperation. It will also identify the conditions under which AI advice improves collective outcomes, and the conditions under which it may instead reinforce defensive or self-interested behaviour.

Experiment 2: AI Social Planning in Networked Common-Pool Resource Games

Aim

Experiment 2 will test when humans cooperate in networked groups, can an AI-designed redistribution mechanism improve cooperation without reducing perceived fairness and autonomy? How to interpret the AI-designed mechanism to make it more acceptable to humans? The goal is to test whether AI-designed mechanisms can improve cooperation in real human groups, and whether interpretable or simplified versions of these mechanisms are more acceptable than black-box policies. And what kinds of recommendations regarding resource allocation, based on social network structures, does it reveal? This experiment asks whether AI-designed allocation mechanisms can improve cooperation in networked common-pool resource games, whether interpretable AI-derived rules are more acceptable than black-box policies, and whether these rules reveal how allocation should depend on individuals' social network structure.

(This experiment builds on my previous work on reinforcement learning in network common-pool resource games and extends it from simulation to human behavioural experiments.)

Design

Participants will be assigned to positions in a network and will repeatedly play a common-pool resource game with their local neighbours. In each round, participants decide how much to contribute to or extract from a local shared resource. A social planner then redistributes resources according to the assigned mechanism. The planner does not control participants' choices, but can shape incentives through redistribution. The AI planner

will be trained using a human-in-the-loop pipeline. First, behavioural data will be collected from human participants playing baseline networked common-pool resource games. Second, behavioural cloning will be used to train virtual human agents that approximate human decision-making as a function of local resource states, previous contributions, neighbours' behaviour, and individual payoff history. Third, a graph neural network-based reinforcement learning planner will be trained in this simulated human environment. Finally, the learned mechanism will be tested with new human participants. The AI planner will be compared with several baseline mechanisms: fairness-oriented equal allocation, contribution-based proportional allocation, or interpolation baseline in [1]. I will also test an interpretable rule extracted from the AI policy, allowing me to compare a black-box AI planner with a simpler AI-derived mechanism.

The primary behavioural outcomes will be average cooperation level, accumulated resources, and fairness across players. After the game, participants will rate each mechanism on fairness, transparency, trust, legitimacy, and willingness to use it again. Where possible, participants will also vote between alternative mechanisms.

Predictions

I predict that AI-designed and AI-derived mechanisms will increase cooperation relative to the no-intervention baseline. However, the interpretable AI-derived rule may be preferred over the black-box AI mechanism, even if the two achieve similar levels of cooperation. This would suggest that human acceptance depends not only on outcome quality, but also on whether the intervention is understandable and normatively acceptable.

A second prediction is that behavioural effectiveness and human satisfaction may diverge. A black-box AI planner may perform better under certain conditions, but its outputs are less robust and predictable to different conditions and unseen environmental states, so user satisfaction may be lower than that of the interpretable AI-derived rule.

A further prediction is that interpretable AI-derived rules may reveal how redistribution should depend on network position. The planner may not treat all individuals identically: instead, it may assign different weights to equality, contribution, and self-protection depending on an individual's degree. For highly connected individuals, relatively egalitarian redistribution may help stabilise cooperation by supporting disadvantaged neighbours and preventing local resource scarcity from spreading. For individuals with intermediate degree, contribution-based redistribution may provide the strongest incentive for continued cooperation. For low-degree individuals, greater self-retention may be necessary to protect them from resource loss and poverty traps. Such findings would turn the AI planner from a purely predictive tool into a source of interpretable hypotheses about how fair and sustainable redistribution should be organised in networked human groups.

Expected Contribution

This project will contribute to the study of AI and cooperation in three ways. First, it will test whether AI-designed mechanisms can promote cooperation in repeated social dilemmas beyond purely simulated environments. Second, it will distinguish between the behavioural effectiveness of a mechanism and its psychological acceptability to human participants. Third, it will identify whether transparency and democratic input make AI interventions more robust, legitimate, and generalisable.

The project therefore addresses not only whether AI can promote cooperation, but also when humans are willing to be guided by AI in collective action problems. This is directly aligned with research on using AI fairly and beneficially to help humans coordinate behaviour and resolve conflicts over shared resources.

Fit with the HIP Lab

The proposed project fits the Human Information Processing Lab's interest in human decision-making, human preferences over AI systems, and the use of AI to promote cooperation. It builds on recent work showing that

AI can help humans find common ground and that deep reinforcement learning can discover mechanisms that promote sustainable behaviour. My background in strategic games, cooperation, reinforcement learning, and computational modelling would allow me to contribute to this programme while extending it toward the psychological mechanisms that determine human acceptance of AI-designed interventions.

References

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